



ashwa RACING

NEWSLETTER - JUNE 2017



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RZX7-H

The race-car designed by the team for the 2017 race season is the efficient and reliable RZX7-H which is a series hybrid prototype, powered by both the motor, Agni-95R and Pulsar 220cc bike engine. The prototype weighs 308 kg, has a space framed chassis and uses a Lithium ion battery with a Battery Management System. It has a Torsen differential and a rear actuated pedal assembly. The suspension system has unequal A-arms with ZF dampers.

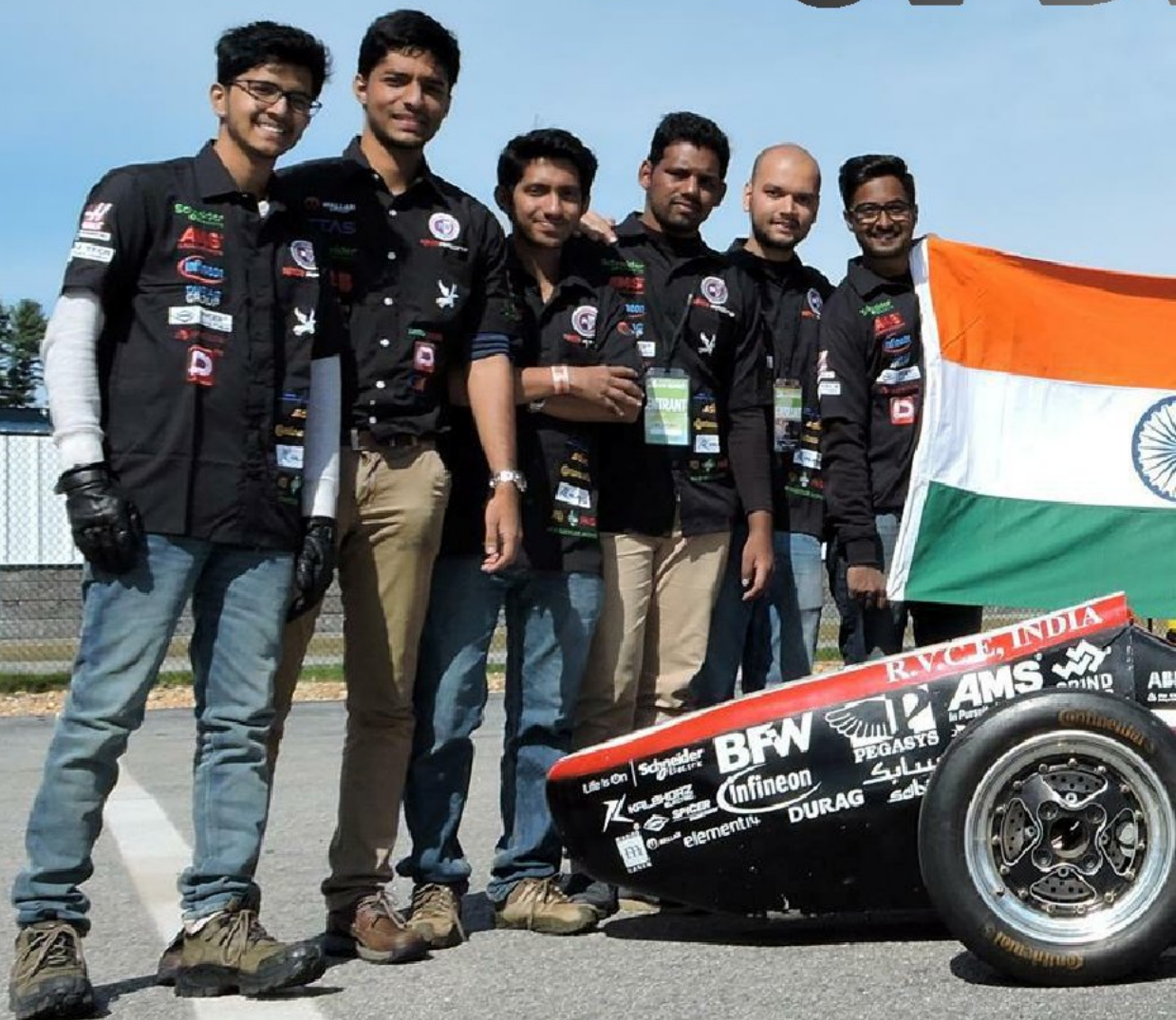
The Pneumatic actuated clutch, an efficient electronic throttle body and a reliable compound drive ensure that the prototype's performance is way ahead of the game.

The main design goal in building this race-car is the performance and reliability. The challenge is to abide by the 35.5 MJ energy limit rule and suitably decide on the architecture to enhance performance.

Off to New Hampshire

On the 27th of April, the team left for New Hampshire to reassemble the RZX7-H which was shipped to the US in March. After spending two days working on the car in the garage of Mr Michael Bouchard (a well-wisher of Ashwa Racing), the team transported the car to New Hampshire Motor Speedway.

Hybrid UPD.



Event DATE



Day

00

Events :

→ PRELIMINARY TECH INSPECTION

Day Zero of the event, which was on 30th April, was an optional day for the Preliminary tech-inspection for teams which wanted to undertake it. Ashwa Racing chose to attend the inspection and by the end of the day, was among the only two teams to have cleared it!

On the same day, the team also had the pleasure of meeting Mr. Dilbagh Gill, the team Principal of Mahindra Formula E Racing and an alumnus of R. V. College of Engineering. The team explained to him the improvements that had been made from the 2016 prototype to which he gave his valuable inputs.

He also presented the team with a scaled down model of the Mahindra Formula E race-car and an envelope containing USD 1000 for the team's yearlong efforts put into the project. Ashwa Racing is indebted to him for all the support he has provided over the years.





Day

01

Events :

MECHANICAL
SCRUTINY

On the next day, the vehicle was inspected to determine if it complies with the mechanical and structural requirements of the rules of the competition. This inspection included examination of the driver's equipment, a test of the emergency shutdown response time and a test of the driver egress time.

The technical inspectors appreciated the team's preparation for the inspection and also mentioned that it was the first time in the history of Formula Hybrid, a team had cleared the Grilling Mechanical Scrutiny in its first attempt.

Day

02

Events :

→ PROJECT
MANAGEMENT
PRESENTATION

→ DESIGN EVENT

→ ELECTRICAL
SCRUTINY

Day two began with the Project Management presentation which evaluates the team's ability to structure and execute a project management plan that helps the team define and meet its goals. The team captain, Sweekruth Shetty and Project Manager, Rakesh H N presented the plan and secured the second place with a score of 94.32 / 100

It was followed by the Design event in which the team convincingly elaborated on the design aspects of the prototype and secured the fifth position with a score of 150.52 / 200

The team also had to sweat blood to clear the dreaded Electrical scrutiny by the end of the day.



Tilt Test



Rain Test



Brakes Test



Autocross



Acceleration

Day

03

Events :

- TILT TABLE TEST
- BRAKES TEST
- RAIN TEST
- ACCELERATION
- AUTOCROSS

On the third day, the team cleared tilt table test wherein the car must be capable of being tilted to a forty-five degree (45°) angle without leaking fluid of any type.

Post tilt, the brakes test had to be completed. After several attempts and optimization of the Brakes and Suspension system for the Brakes test track, the team was able to successfully clear the test.

As the track was declared damp, the Rain test had to be taken on, which was cleared just in time to attempt the dynamic events.

First was the Acceleration event, which was broken down into two subevents: Electric mode & Hybrid mode. The team secured the first position in the Electric mode and second in the Hybrid mode.

Next was Autocross, which evaluates the car's maneuverability and handling qualities on a tight course. The autocross course combines the performance features of acceleration, braking, and cornering into one event. The team stood second in the Autocross event.

Day

04

Events :

— ENDURANCE

This day was dedicated for the Endurance event - the toughest and highest scoring dynamic event which is designed to evaluate the vehicle's overall performance, reliability and efficiency.

The endurance distance is approximately 44km (27.3 miles) comprised of four 11 km (6.84 mile) segments.

Formula Hybrid rewards the team that can cover a designated distance on a fixed amount of energy in the least amount of time.

The team completed a partial Endurance, after which the car shut down due to a fault in the batteries detected by our Battery Management System.



Endurance

VICTORY!!



At the end of the 5 day ordeal, Ashwa Racing was placed at 2nd overall on a Global scale; securing the first podium finish for the team in its 15 years of history and the best event ranking for any FSAE team from the country!

The team bettered the previous year's results by climbing the score ladder by two steps.

With an unforgettable experience and learning, the team returned to its motherland, victorious.



FINAL STANDINGS

FORMULA HYBRID 2017



Static Events

1. Project Management Plan - **2nd**
2. Design Presentation - **5th**

Dynamic Events

1. Acceleration
 1. Restricted - **1st**
 2. Unrestricted- **2nd**
2. Autocross - **2nd**
3. Endurance - **2nd**

Overall 2nd

Fun Amidst the Frenzy



All the way from Chandigarh

Procurement of Steel Tubes



Procurement of Steel Tubes from BHARATIYA MANUFACTURING INDUSTRIES, Chandigarh in the month of May has kick started the manufacturing phase of the upcoming race season.

Grade:
AISI 1018

Type:
Seamless

Minimum carbon content - 0.10%

Young's Modulus - 200 GPa

Yield Strength - 305 MPa

Ultimate Strength - 365 MPa

Ashwa Racing would like to thank BHARATIYA MANUFACTURING INDUSTRIES, Chandigarh for supporting the team's endeavour and providing the required Steel Tubes at a 22% discounted price.

LET'S Football

Bengaluru Football Club is an Indian professional football club based in Bengaluru, Karnataka. The club participates in the I-League, India's top-tier football league.

Ashwa Racing is an ardent follower of BFC. The team would like to wish BFC all the very best for their upcoming league.



Bengaluru FC

@bengalurufc

The BFC family wishes @ashwaracing, who will participate in the upcoming Formula SAE Italy event in July, all the very best!



Ashwa Racing would like to thank the **BFC FAMILY** for giving the team an opportunity to display the RZX7-C prototype at the Kanteerva stadium during a match between BFC and Churchill Brothers amidst a young, enthusiastic crowd who were attracted by the stylish RZX7-C and were highly appreciative of the team's work.



SPONSOR FOCUS



CONTINENTAL has provided us with dry and wet tyres for the 2017 race season. They have also been very supportive by helping us monetarily and have backed the team with constant encouragement and guidance right from its inception. We are extremely thankful to CONTINENTAL for their continued support.

MOTUL, an industry specialist in high performance motor oils and industry lubricants helps us meet our oil and lubricant requirements. We thank MOTUL for the provision of all required fluids and their continued support towards the project.



ROYAL BROTHERS is India's Largest and South India's First RTO Authorized Self-Ride Bike Rental Service Provider, operating in 6 different cities. Ashwa Racing would like to thank ROYAL BROTHERS for making our journeys less cumbersome, with their rental bikes.

AMF is the leading manufacturer in Europe, supplying over 5,000 different products from the fields of clamping, hand tools and locks. We are proud to have AMF in our family of sponsors.



WRAPCRAFT is an automotive styling and vinyl customization company. Wrapcraft has shown immense growth and made a name for itself amongst the automotive crowd. The team whole heartedly welcomes WRAPCRAFT to our family of sponsors.

AUDIT - MAY 2017



R. V. COLLEGE OF ENGINEERING

अश्वरACING

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+91 81479 67789



R. V. COLLEGE OF ENGINEERING / **अश्वरACING**

Accounts for the period from 1-05-2017 to 31-05-2017

RECIEPTS	AMOUNT
Opening Balance in Kotak Mahindra Bank RVCE Branch	
SB A/c 136010046665	₹ 1,445,981.82
SB A/c 136010058514	₹ 1,037,863.26
SB A/c 136010060793	₹ 352,994.93
new hampshire balance money	₹ 14,947.00
37H,28C excess money	₹ 1,480.00
cash in hand	₹ 3,960.00
italy flight ticket	₹ 296,000.00
X6C batch seed money	₹ 4,353.00
X7C batch seed money	₹ 9,004.00
sponsorship money	₹ 289,000.00
insurance for X7C	₹ 8,944.00
TOTAL	₹ 3,464,528.01

PAYMENTS BY	AMOUNT
Powertrain & Smp	₹ 54,731.00
Electrical	₹ 92.00
Miscellaneous	₹ 11,854.00
Brakes & Logistics	₹ 229,270.00
Chassis & Suspension	₹ 43,912.00
SMS & Wire Charges	₹ 431.28
Cash In Hand	₹ 1,460.00
Closing balance in Kotak Mahindra Bank , RVCE Branch	
SB A/c 136010046665	₹ 1,924,568.68
SB A/c 136010058514	₹ 677,242.12
SB A/c 136010060793	₹ 520,966.93
TOTAL	₹ 3,464,528.01

Utkarsh

Utkarsh Agrawal
Finance Team

अश्वरACING

R.V. College of engineering

ALUMNUS OF THE MONTH



Gopika Hemachander, fondly nicknamed Gopi, was a part of the steering and finance module of the team from 2009 until 2012. She helped to set up and understand the steering geometry developed during her time and brought in new perspectives to the subsystem. The learnings from the RZX2 car that she worked on, has helped set a benchmark for steering in Ashwa that has had a strong influence on the future cars including the current prototype.

She is an extremely hard-working person who is very passionate about the team and its future.

She worked on the cost report and presented the same at the event.

She was instrumental in helping the team secure the 2nd position in the cost event at the FSAE Italy 2012.

She is a kind hearted person who can always be counted on and has the unique quality of always having a bright and cheerful smile. Her infectious smile spreads a positive vibe and keeps the team's morale high.

Her passion for the formula student environment has led her to be a part of the Formula Bharat organising committee and also volunteer as a cost judge during the event. We wish her all the best in all her future endeavours and are confident that the team can continue to rely on her for guidance and support for many years to come.

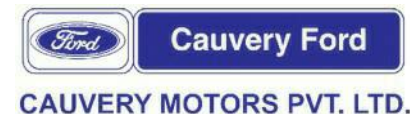
Thank you Gopi for all the help and support.



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Continental 



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SUN PRECISION TOOLS

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sebc

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